AI/ML-Based Monitoring of SOP Variations in Optical Networks

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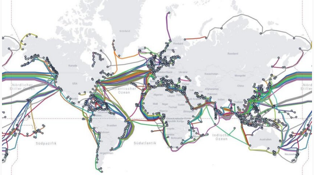
Importance of Optical Networks and Emerging Threats

- Optical fiber networks carry over 95% of global internet traffic. But this backbone is **physically vulnerable**:
  - Eavesdropping via subtle bends or taps
  - Service degradation from construction, mechanical stress or vibration
- Conventional monitoring tools like OTDR and DFOS are:
  - Expensive and hardware-heavy
  - Blind to short, subtle, or overlapping disturbances
- **Research Question: How can we enable intelligent, low-cost, and always-on fiber sensing technique?**




'Malicious' Attacks on French Fiber Networks Cause Internet Outages

By Benoit Berthelot | April 28, 2022

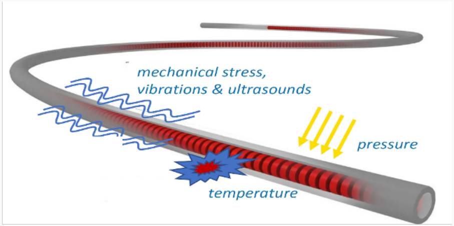

Red Sea cables have been damaged, disrupting internet traffic

By Hanna Ziady, CNN

3 minute read · Updated 11:25 AM EST, Mon March 4, 2024

State of Polarization (SOP): From Optical Property to Sensing Engine

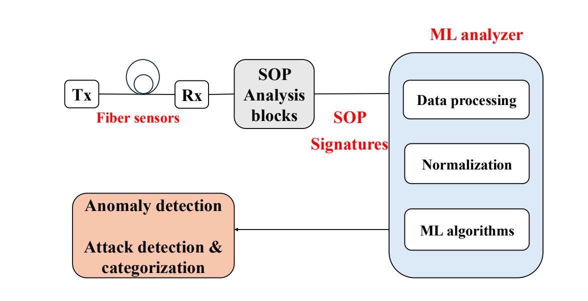
- **Polarization in optical fibers**
  - SOP is changed by external disturbances.
  - SOP variations as indicators and dual functionality role of fibers



- **Our Solution**:
  - Turn SOP into a digital fingerprint for fiber sensing and develop AI/ML techniques to detect anomalies from the unique *SOP signatures* for each event.

AI/ML-Based Analysis of SOP Variations

- **Why ML-Driven SOP monitoring?**
  - Traditional thresholding fails to capture subtle, compound, or emerging events.
  - ML enables scalable and automated anomaly detection, event classification, and feature extraction from dynamic SOP patterns.
  - Can detect the previously unseen attack types.
- **Proposed framework**



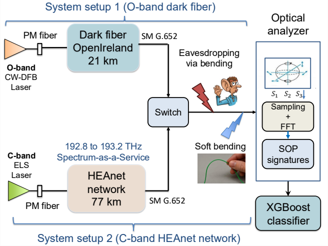
Research Contributions: From Detection to Cognitive Monitoring

- **Data Collection:**
  - Multi-attack datasets across 3 cable types (bare, indoor, FOCS) from lab and OpenIreland testbeds.
- **Supervised Learning Achievements**:
  - Detection and categorization of normal and harmful events.
  - Achieved up to **99%** accuracy for experimental and up to **90%** accuracy for real-world noisy environments (OFC 2024-2025, ECOC 2024, ICTON 2024-2025).
- **Deep Learning Extension**:
  - Short-link (300 m) and long-link (21 km) event detection and categorization over the OpenIreland testbed (JLT 2025).

Installation	**Training**	**Validation**	**Testing**
Short-link	99.81	98.90	98.57
Long-link	95.28	92.77	92.26

Latest Advances: Real-World SOP Sensing Over HEAnet Infrastructure

- **Main Problem:** Generalization of ML models across spectral bands and fiber links (ECOC 2025).



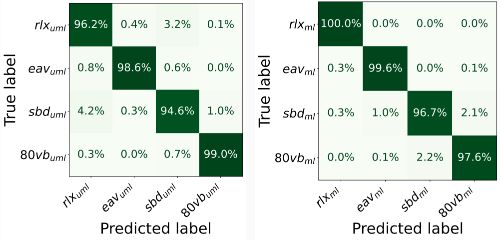
Experimental Setup

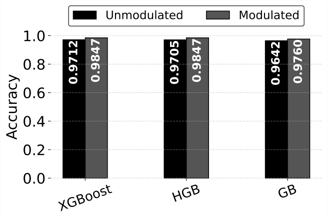
Classification accuracy (Acc) for the different training/testing scenarios.

Scenario	Training	Testing	Acc.
S_1	System1	System1	88.85%
S_2	System2	System2	98.63%
S_3	System1	System2	8.11%
S_4	System2	System1	60.59%
S_5	System1+2	System1+2	91.11%

Latest Advances: Real-World SOP Sensing Over HEAnet Infrastructure

- **Main Problem:** Impact of signal modulation on SOP signatures and ML classification (JLT 2025).





Thank You For Your Attention.

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